

(a) Data sources

Landsat surface reflectance

USGS · Collection 2, Level 2 · 30 m

LANDSAT/LT05/C02/T1_L2

LANDSAT/LE07/C02/T1_L2

LANDSAT/LC08/C02/T1_L2

Built-up surface, population

European Commission JRC · 100 m

GHS-BUILT-S R2023A

GHS-POP R2023A

Roads, terrain, water

OpenStreetMap (Overpass API)

USGS/SRTMGL1_003

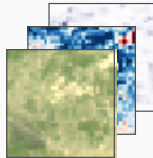
GOOGLE/DYNAMICWORLD/V1

(b) Pre-processing



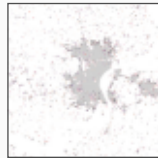
Oct–Mar median composites
every 5 years, 1995–2020
QA_PIXEL cloud mask
TM/ETM+ → OLI (Roy 2016)
125,925 cells of 100 m
urban: built-up ≥ 0.20

(c) Input tensor



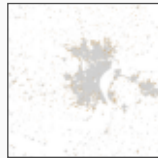
$45 \times 32 \times 32$ patches
10 bands, indices at t
10 at $t - 5$, 10 changes
15 place drivers
stride 8, z-scored

(d) Dual-path CNN



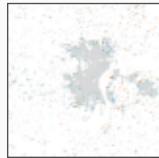
115,345 parameters
focal + 0.5 Dice loss
5 seeds $\times$ 4 rotations
→ *suitability* S

(e) CA allocation



$s = 0.65 S + 0.35 n_{300}$
8 iterations
demand = observed count
→ *new urban cells*

(f) Evaluation



test 2015 → 2020, unseen
reference: GHS-BUILT-S
FoM 0.151 · AUC 0.935
■ hit ■ miss ■ false alarm